

U.S. Competitiveness

Where Do We Stand? What Do We Do Now?

By Richard Elkus Jr.

Gordon Moore, a former co-founder of both Fairchild Semiconductor and Intel, famously predicted in 1965 that the number of transistors in an integrated circuit could be doubled every 18 months to two years, thereby increasing computational power of the device exponentially. At the same time, the relative cost of that computing power could be halved. Moore's Law turned out to be amazingly accurate. The implications were astounding. Those products and markets affected by this exponential increase in computational power tied to the development of integrated circuits (semiconductors) produced technological advances that heretofore could not have been imagined. As a result, the products and markets influenced by this increased ability to process information expanded exponentially.

But there was a downside. The investment in monetary and human capital required to achieve the results of Moore's Law goes up exponentially as well. These costs were to be borne by the exponential growth in new products designed to replace existing products and markets. But if a nation is not able, for whatever reason, to maintain and grow its required technological infrastructure, and therefore is not able to participate in these new growth markets, the cost to try and catch up later quickly becomes overwhelming to its economy. As such, Moore's Law (which became a guiding factor in the investment in the semiconductor industry) tied a nation's investment history to its future.

Once a nation begins to lose the infrastructure that supports its strategic technological development, its future and the independence of its people are in jeopardy. Most countries are not able to support the investment in human capital and monetary resources needed to sustain sufficient technological infrastructure if it is financed solely from its own resources. Hence strategic alliances to ensure the technological future of nations become essential to maintain economic viability as well as political stability and security.

Without dependable strategic alliances or the internal capability and resources necessary to maintain integrated technological advancement commensurate with the competition, most nations will fail to maintain and finally lose their future competitiveness and independence. This set of circumstances is a major problem for the United States today. The foundation of the U.S. technological base, which is

necessary for it to meet its goals and objectives of leadership among nations and its desire to maintain itself as a vital strategic partner to its allies, is being threatened from within. The United States, the beacon of democracy, is in trouble. Though the United States possesses a wealth of ideas and creativity, it is rapidly losing the technological infrastructure necessary for competitive leadership. If this trajectory is not changed soon, the cost in human capital and monetary investment may become more than its population, and therefore its political system, can bear. The U.S. political system, based in part on its economic growth and stability, which has been the source of the United States' rise to prominence, may become permanently damaged.

Assuming computing capacity has doubled every 18 months to two years as predicted by Moore's Law, computational power today is approximately 2 billion times what it was when the integrated circuit was invented. (Moreover, in the coming years, more advanced computational software, and hardware, applicable to quantum computing, may cut the projected rate of doubling computing power in half.) As a result, technological advancements have dramatically improved products and services across the total spectrum of society. The technological possibilities of today could not have been imagined five years ago, let alone 60 years ago. Today's products and services have completely altered the political, economic, and military landscape of the world. But the capital investment needed to establish the technological infrastructure necessary to maintain Moore's Law has also gone up exponentially—capital investment in technological infrastructure that cost millions of dollars 60 years ago costs many billions today.

At one time, the United States dominated key products and markets the world over.

After World War II and into the 1950s and 60s the United States dominated most key strategic products and markets, including integrated phone systems, telephones, computers, consumer electronic products in general, mass-produced automobiles, ships, and aircraft. The best laboratories that fostered the technological advances inherent in these products and associated services, such as Bell Laboratories within AT&T and Sarnoff Laboratories within RCA, resided in the United States. Bell Labs contributed groundbreaking technological inventions, including data networking, the transistor, cellular telephone technology, laser technology, and digital signal processing. Sarnoff Laboratories provided the United States with monochrome compatible electronic color television, liquid crystal displays, and thin film technologies applicable to television camera technologies and later the development of semiconductors. More importantly, as production of products emanating from these inventions became more sophisticated, and automated, the manufacturing of the components, as well as the end products, was largely dominated by U.S. companies. These manufacturing entities included machine shops, sheet metal shops, and product assembly operations, all closely integrated with the manufacturer of the product.

As the technological sophistication of these products and markets increased, they became more and more dependent upon the collection, processing, and distribution of information. As a result, the ability to store and process information became an area of concentration for all business activity. Information that was stored on film, records, tape, punch cards, and other electromechanical devices was being developed and produced to support and advance the state of the art of all products. Many companies benefited from the advancement of information storage and processing, including General Electric (GE), IBM, RCA, Motorola, Zenith, Sylvania, Kodak, Ampex, AT&T (Western Electric), Ford, Chrysler, General Motors, and a myriad of other companies.

Starting in the late 1940s, from Bell Laboratories at AT&T, and then followed up from Intel, and Texas Instruments in the late 1950s came the development of the transistor and the integrated circuit. Immediately, because of the semiconductor's ability to facilitate the processing of information, the

potential performance and sophistication of all products increased in terms of form, fit, and function. Moreover, in various ways, they could all be interconnected, as information was now transferable from one product to another.

Then, the United States began to lose its competitive position.

As technology was advancing, the United States, feeling impervious to competition from other nations and coupled with the quarterly reporting of financial results, adopted a short-term planning matrix: Stars, Cash Cows, and Dogs. Simply stated, you invest in the stars, you milk the cash cows, and you sell off the dogs. Quarterly profits and revenue growth became the name of the game. Other nations, particularly in Asia (and initially in Japan) saw a way to join in the technological growth of the economic system. Concentrating less on short-term profits, they offered to put up the capital and take over production of subassemblies and other low-cost, low-margin products that were part of this infrastructure but of low profitability to the U.S. industry. One of the key industries they concentrated on was consumer electronics.

As performance improved and the cost of these consumer products and subassemblies from Asia (principally Japan) fell, demand for Asian production grew. This led to increasing sophistication in high-volume manufacturing for parts and assemblies in Asia. As manufacturing expertise and automation increased, the cost of establishing a new manufacturing facility also started to grow exponentially, making reentry into those manufacturing areas by U.S. companies less desirable or simply too costly.

But there was another cost to the loss of these Asian manufactured products oriented toward consumer electronics. Consumer products rapidly became the largest user of semiconductor devices of any products produced at that time. As such, in accordance with Moore's Law, the market for semiconductors in Asia began to support significant Asian investment in the semiconductor industry that enabled them to manufacture subassemblies and products that began to exceed the sophistication and performance of U.S. manufacturers.

With advances in information processing, the parts, assemblies, and end-use products became highly integrated. For example, the television set, the tape deck, the tape cartridge, and power supply of a video cassette recorder (VCR) morphed into a cell phone that became millions of times more powerful, at a fraction of the cost, with technological versatility that could not be duplicated by the VCR. As these new, highly integrated products grew in popularity and as they became integrated and more versatile, Asian competition began to dominate not only the manufacturing of key subassemblies for U.S. end-use products, but they also began to compete in the end-use market directly. As Asian manufacturers began to control the design and manufacturing of subassemblies, they saw the obvious benefit of integrating those subassemblies into the design of the end-use product, enabled largely by using semiconductors. The result was almost immediate. These subassembly manufacturers began to take over U.S. markets for end use-products as well. Major markets affected included automobiles, VCRs, telephones, cameras, televisions, radios, the display industry, and consumer electronics in general. These products, as they grew in sophistication, began to become integrated functionally because of the adoption of semiconductors. The Japanese automobile industry widely adopted integrated circuits years before the U.S. automobile industry and as a result were producing car entertainment systems, automatic wipers, electronic locks, and other controls well before they were available in U.S. vehicles.

The result has been catastrophic for the U.S. industry as it lost much of its manufacturing expertise. RCA, one of the largest consumer electronic manufactures in the world, was bought by GE and subsequently sold to foreign corporations or liquidated. Motorola, once a major U.S. consumer electronics company and semiconductor manufacturer, who first invented the cell phone, is a fraction of its former size. After

its troubled business was split up, the bulk of its consumer electronics operations was sold to Lenovo in China, and its semiconductor business was ultimately bought by NXP in Europe. Zenith is gone, and its fundamental technology developed for High-Definition Television was sold to LG Corporation in South Korea. Sylvania is gone. Kodak, who first invented the digital camera, but decided to stay with Kodachrome film because of its huge profitability, went bankrupt. Ampex, who invented video recording, went bankrupt. AT&T has been broken up and is a fraction of its original size and reach. Western Electric, one of the world's largest manufacturers of phones and phone systems, is gone. IBM, which dominated the PC business that paved the way for Microsoft and the computer business in general, and at one time was dominant in semiconductors, nearly went bankrupt. IBM sold its PC business to Lenovo in China, and as of today, has had basically no growth for the last 25 years. Bell Labs is gone. Sarnoff Labs is gone. GE, once the most valuable U.S. manufacturing corporation, is a shadow of its former self. Apple, like many other U.S. manufacturers of consumer electronic devices, purchases a major portion of their integrated circuits from the Taiwan Semiconductor Manufacturing Company (TSMC) and assemblies from Foxconn in Taiwan, China, and elsewhere in Asia.

As a result of the loss of these domestic manufacturing relationships and therefore general demand for U.S. semiconductor products, coupled with the huge capital investments required to remain a fabricator of advanced semiconductor devices, U.S. dominance of the fabrication of integrated circuits (semiconductors)—devices at the core of today's technological advancements—is a fraction of its original size. The United States once represented over half the market—today, it fabricates less than 12 percent.

For all practical purposes, the United States no longer manufactures displays (which depend on their functionality from the use of semiconductors). Beyond just entertainment, displays are fundamental to the analysis and utilization of information.

Japan became formidable in automotive production as it introduced fully automated plants. Ultimately, General Motors, after spending 90 billion dollars trying to replicate the automation of Japan's automobile factories and then having to face the financial crisis of 2007-2008, went bankrupt. Chrysler was bought by Fiat.

And there is one more very important point. As competitive technological infrastructure, originally dominated by the U.S. industry, shifted to Japan, and then to other Asian nations, most of it never came back to the United States. With some exceptions, the United States has continued to lose portions of its technological infrastructure and therefore has had to outsource more and more of those strategic products to foreign companies.

As individual products and then markets become integrated, largely with the advancement of information processing, so do the businesses that control them. Look at Amazon. It started distributing books over the internet; then it took on other products, then it took on the retailing and distribution of almost all other products. But that required fast and reliable distribution. This required the development of a major airfreight business as well as significant investment in ground transportation. These businesses in turn supported and were supported by the internal development of cloud computing, large and sustained development of information processing capabilities including payment systems, and international distribution.

This same broad capability is available to other nations (consider China's Alibaba) that have a similar broad technological infrastructure (and in the case of China, a population four times larger than the United States). But an Amazon-like company in a country such as China will have a potentially significant advantage. It will represent a nation that is able to develop and manufacture a significant number of the products that are bought and sold by these integrated distribution companies, thus broadening its control

over the supply chain. Control of both manufacturing and distribution of a massive array of products and services in this manner could lead a company such as Alibaba to become a competitive juggernaut.

And there is another factor to consider. The integration of these products and markets as information processing becomes more sophisticated gives companies (friendly and unfriendly) the ability to connect their systems with others, whether that connection is wanted or not. In other words, advancements in technological infrastructure will enable individuals, companies, and nations to influence others (both as partners and enemies) in ways never considered before.

What's the answer?

There is no simple way to control the advancement of technology and the effects it has on society. But one thing is certain: if a nation such as the United States does not understand the problem and therefore comes up with solutions that do not address the fundamental issues associated with a declining technological infrastructure, its future is going to be bleak at best. The technological infrastructure of the United States must be competitively equal if not superior to all other nations. And the United States certainly should avoid dependence on a supply chain that may be controlled by a potentially unfriendly competitor.

Right now, Taiwan is a flashpoint between the United States and China. That territory controls, at very advanced levels, technological infrastructure that is fundamental to the competitiveness of the United States and China. As this paper is being written, both nations want to limit the industrial base of Taiwan from providing the other nation with a significant competitive advantage: whether it be political, economic, or military. The danger is that if either nation thinks the other's aggressive actions will give the aggressor a major technological advantage, there will be a strong motivation to limit or eliminate that advantage. This could cause a miscalculation by either party in their desire to keep the playing field level. And that could lead to a military conflict.

Today, an interruption in the relationship between China and Taiwan would prove to be more difficult and strategically significant for China than a break in relations between Taiwan and the United States would be for the United States. Of course, a loss of access to Taiwan's technological infrastructure could still pose a major problem for the United States. Taiwan provides vital semiconductor products and assemblies for manufacturers of cell phones and a myriad of other consumer electronics no longer produced in the United States. Taiwan is a very large manufacturer of PCs, televisions, consumer electronics in general, and other communication equipment. Without Taiwan, Apple, AMD, Qualcomm, Nvidia, Texas Instruments, and HP would be in trouble, as would Intel, to an extent, with implications for their respective customers. TSMC, a bulwark of the Taiwanese economy and the largest foundry for semiconductors in the world, currently has the most advanced fabrication process. So, losing Taiwan and its key manufacturers would be a very severe blow to the U.S. industry. If Taiwan were, for whatever reason, unable to supply products to both the United States and China, China would suffer the most. China is behind the United States in semiconductor technology, software technology, and the production of advanced semiconductor manufacturing equipment. On the other hand, as China's technological infrastructure advances, especially based on its current relationship with Taiwan, China will become less and less dependent on direct support from Taiwan (one of China's principal objectives). In time, China will have less need for companies such as TSMC as it transfers, builds, or in some way controls a similar technological infrastructure on the mainland. This change will not favor the United States. Under this scenario, Taiwan will become more important to the United States than to China, providing China with significant additional leverage over a major piece of U.S. technological infrastructure.

The United States does have a significant competitive lead in software today. But software does not exist in a vacuum. Semiconductors, computers, displays, phones, and other communications hardware are essential if software is to have value. The United States also leads in the manufacturing of semiconductor fabrication equipment. This gives the United States an edge as an enabler in the construction of semiconductor fabrication facilities. But it is difficult to control the utilization of that equipment when it is used to produce semiconductors manufactured by others in foreign fabrication facilities that may be more advanced than U.S. ones. All things considered, the United States must be able to deliver technologically competitive products and services without the fear of a major disruption in its supply chain.

Recapitalizing the U.S. industrial base and getting back into the businesses that have been lost is very important and ultimately essential. But the time it will take, and the amount of capital that will be required to rebuild that technological infrastructure may not overcome the damage that will occur if the supply chain that the United States depends on today is interrupted or broken. To replace, by itself, the technological infrastructure that the United States has lost will cost hundreds of billions of dollars, require the re-creation of a massive and highly trained labor force, and invite huge execution risks. At the same time, computational capability is doubling every two years, as is the technological infrastructure of the competition.

The United States needs to immediately fortify and diversify its supply chain that supports the strategic technological infrastructure it has lost. The expansion and diversification of its current supply chain must include nations truly allied with the United States and reasonably free from potential impediments that could significantly damage the alliance. It must also be able to provide the United States with strategic end-use products and components produced by manufacturers that have the research capability to keep their technological infrastructure current. Those manufacturers must possess the production capacity to meet the demands of the United States. The main purpose of this strategy of supply chain diversification is to enable the United States to rapidly reduce its dependence on Taiwan.

The United States needs to tap into, where possible, a resource base that already exists and which can provide immediate results. It is essential that the United States keeps in mind the precepts of Moore's Law. In two years, computing power may be twice as powerful as today. Moreover, a single state-of-the-art semiconductor fabrication facility can cost as much as \$30 billion. And this does not include the research investment necessary to build it or keep it current. (A research capability of this magnitude depends upon trained and available human resources that may or may not be available in time.)

There are two nations in Asia that can be very helpful in this situation: South Korea and Japan. Both depend upon the United States as a strategic partner and understand the competitive problem that each of the three nations faces. The United States needs to seek out and develop partnerships with other nations that possess a technological infrastructure that offers the United States the opportunity to reduce its dependence on Taiwan in a relatively short period of time. Much of the technological infrastructure, including the manufacture of semiconductors (principally in South Korea), exists in South Korea and Japan and is competitive in today's market. Both South Korea and Japan have technological resources and production that can replicate what the United States has lost in its technological infrastructure. A strategic alliance with these two countries can offer an alternative supply chain that reduces U.S. dependence on Taiwan.

By developing these partnerships for strategic reasons rather than a simple economic rationale, the United States can reduce Taiwan's significance as a flashpoint with China without forcing or opening the door to a major confrontation.

This should buy the United States time to reinforce its strategic industrial base. The goal of reestablishing U.S. technological infrastructure is essential to the maintenance of its competitiveness as well as its ability

to innovate and commercialize state-of-the-art products and markets. This level of competitiveness is as important for U.S. allies as it is to itself.

There are many tax changes, funding incentives, and laws that can encourage U.S. reinvestment in the United States. The short-term concept of stars, cash cows, and dogs must end. But all of this takes time, requires very large financial and human resource investments, and carries significant execution risk.

Therefore, the United States must diversify, expand, and fortify its strategic supply chain by supporting alternative foreign procurement pathways and alliances now.

But beware: Both Japan and South Korea have been expanding their trade and investments with China for some time. This obviously has increased the leverage that China has over these countries. Hence it is of vital importance that the recommended trade negotiations be part of a well-thought-out series of discussions that include both the respective governments involved as well as related corporate leadership. These negotiations should be a part of a strategic plan designed to further the political as well as economic goals and objectives of the United States, Japan, and South Korea, with the expressed purpose of reducing dependence on Taiwan as well as strengthening the competitive position of the alliance. ■

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This report is made possible by general support to CSIS. No direct sponsorship contributed to this report.

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